

TASK 1 – MACHINE LEARNING CLASSIFICATION PROJECT REPORT

Customer Churn Prediction Using Machine Learning

Submitted By

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Task: Task 1 – ML Classification Project

1. Introduction

Machine Learning classification is a supervised learning technique used to predict categorical outcomes based on historical data. Classification models are widely used in industries for customer retention, spam filtering, fraud detection, healthcare diagnosis, and recommendation systems.

The objective of this project is to build a machine learning classification model that predicts whether a customer is likely to leave a telecom service provider. Predicting customer churn helps organizations identify customers at risk of leaving and take proactive measures to improve customer retention.

In this project, the Telco Customer Churn dataset was used to train and evaluate classification models. The dataset was preprocessed by handling missing values, removing irrelevant features, and encoding categorical variables. Two machine learning algorithms, Logistic Regression and Random Forest Classifier, were implemented and compared using various evaluation metrics. The model performance was assessed using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Cross-Validation scores.

2. Dataset Information

Item	Details
Dataset Name	Telco Customer Churn Dataset
Source	Kaggle
Target Variable	Churn
Total Records	7043

```
[0] ✓ 4s
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   customerID            7043 non-null   object 
 1   gender                7043 non-null   object 
 2   SeniorCitizen         7043 non-null   int64  
 3   Partner               7043 non-null   object 
 4   Dependents            7043 non-null   object 
 5   tenure                7043 non-null   int64  
 6   PhoneService          7043 non-null   object 
 7   MultipleLines         7043 non-null   object 
 8   InternetService       7043 non-null   object 
 9   OnlineSecurity        7043 non-null   object 
10   OnlineBackup          7043 non-null   object 
11   DeviceProtection      7043 non-null   object 
12   TechSupport           7043 non-null   object 
13   StreamingTV           7043 non-null   object 
14   StreamingMovies       7043 non-null   object 
15   Contract              7043 non-null   object 
16   PaperlessBilling      7043 non-null   object 
17   PaymentMethod         7043 non-null   object 
18   MonthlyCharges        7043 non-null   float64 
19   TotalCharges          7043 non-null   object 
20   Churn                 7043 non-null   object
```

Dataset Description

The dataset contains information about telecom customers, including demographic details, account information, subscribed services, and whether the customer has churned.

Target Variable

Churn

- 1 = Customer left the company
- 0 = Customer stayed with the company

```
[5] ✓ 1s
df.head()
5 rows x 21 columns
customerID  gender  SeniorCitizen  Partner  Dependents  tenure  PhoneService  MultipleLines  InternetService  OnlineSecurity  ...  DevicePr
0    7590-VHVEG  Female           0      Yes         No        1         No        No phone service    DSL              No      ...
1    5575-GNVDE  Male           0      No          No        34         Yes         No              DSL              Yes      ...
2    3668-QPYBK  Male           0      No          No        2         Yes         No              DSL              Yes      ...
3    7795-CFOCW  Male           0      No          No        45         No        No phone service    DSL              Yes      ...
4    9237-HQITU  Female          0      No          No        2         Yes         No        Fiber optic     No      ...
5 rows x 21 columns
```

3. Data Preprocessing

Data preprocessing was performed to improve data quality and prepare the dataset for machine learning algorithms.

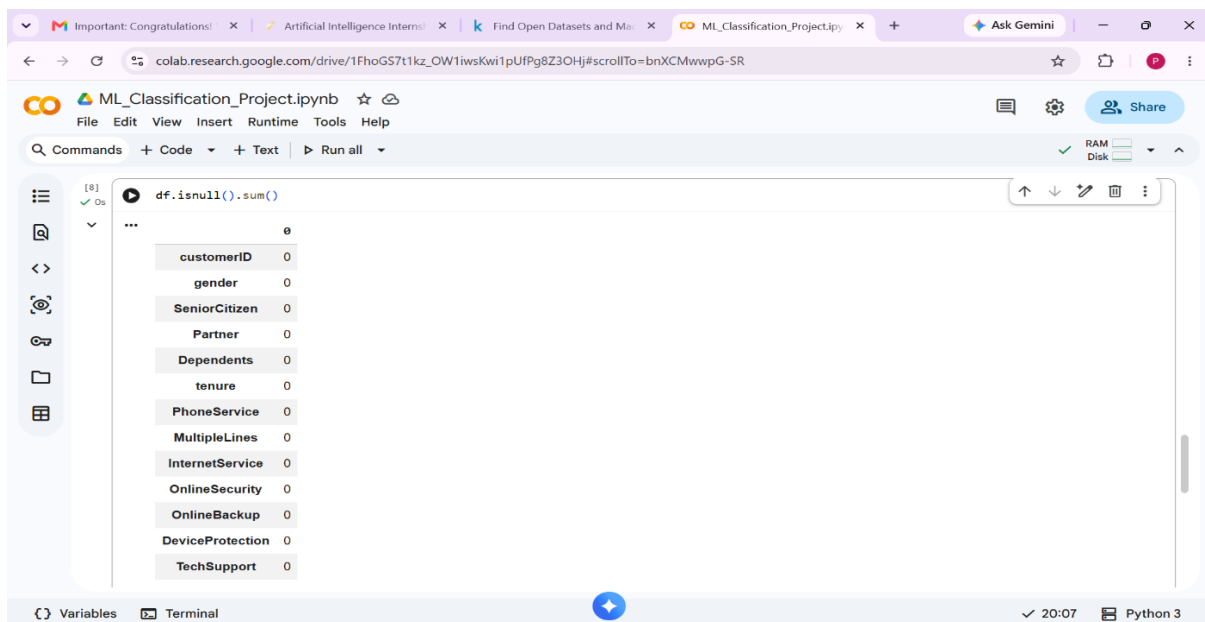
Steps Performed:

1. Handling Missing Values

The dataset was checked for missing values using:

```
df.isnull().sum()
```

Missing values in the TotalCharges column were replaced using the median value.



2. Removing Unnecessary Features

The customerID column was removed because it does not contribute to prediction.

3. Encoding Categorical Variables

Categorical features were converted into numerical values using Label Encoding.

4. Feature and Target Separation

The target variable Churn was separated from the input features.

5. Train-Test Split

The dataset was split into:

- Training Data: 80%
- Testing Data: 20%

4. Machine Learning Models Used

Two classification algorithms were implemented and compared.

4.1 Logistic Regression

Logistic Regression is a supervised learning algorithm used for binary classification problems. It predicts the probability of a customer belonging to a particular class.

Advantages:

- Simple and interpretable
- Fast training
- Effective for binary classification

4.2 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy.

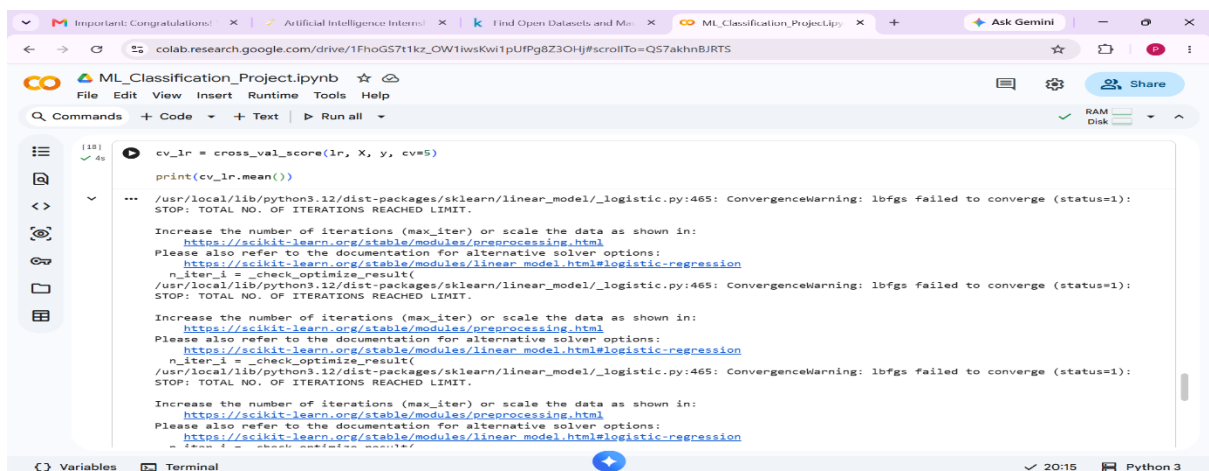
Advantages:

- Handles complex relationships
- Reduces overfitting
- Provides high predictive performance

5. Cross Validation

To evaluate model stability and generalization, 5-Fold Cross Validation was performed.

5.1 Logistic Regression Cross Validation Score



```
cv_lr = cross_val_score(lr, X, y, cv=5)
print(cv_lr.mean())

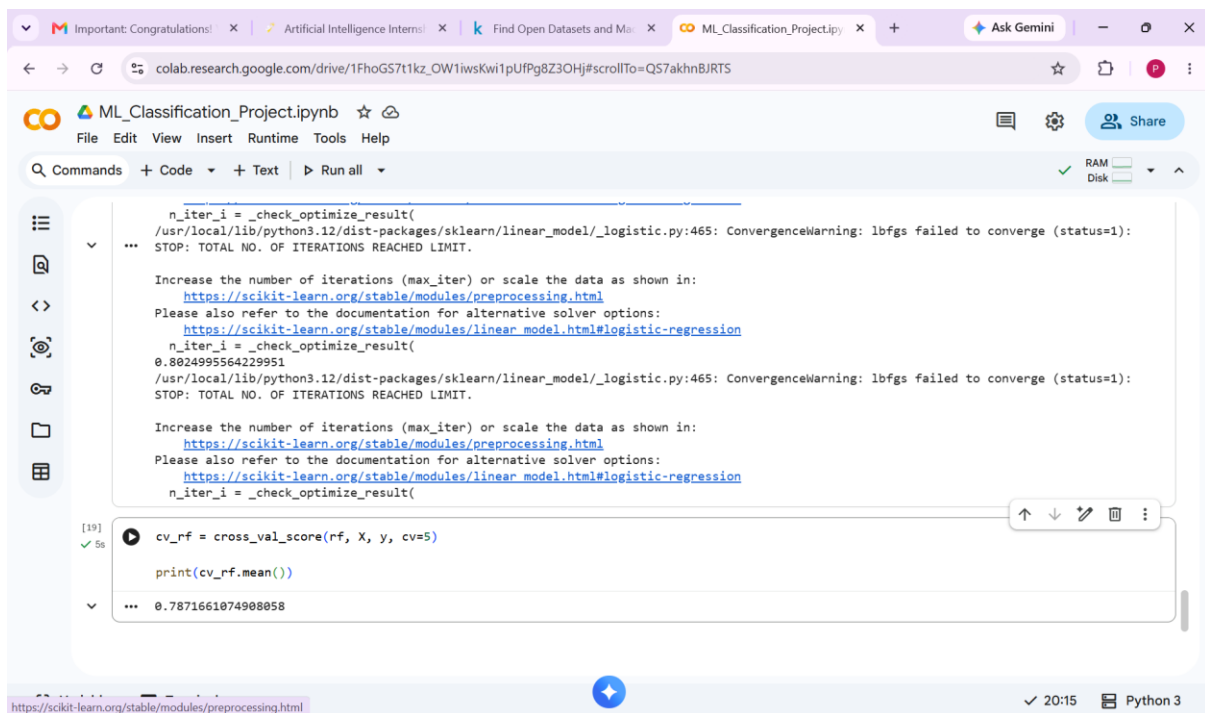
...
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear\_model.html#logistic-regression
n_iter_1 = _check_optimize_result(
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear\_model.html#logistic-regression
n_iter_2 = _check_optimize_result(
n_iter_3 = _check_optimize_result(
n_iter_4 = _check_optimize_result(
n_iter_5 = _check_optimize_result(
cv_lr.mean()
0.8025
```

Mean CV Score = 80.25%

5.2 Random Forest Cross Validation Score



```
n_iter_i = _check_optimize_result(
...
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear\_model.html#logistic-regression
n_iter_i = _check_optimize_result(
0.8024995564229951
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
https://scikit-learn.org/stable/modules/linear\_model.html#logistic-regression
n_iter_i = _check_optimize_result(

[19] cv_rf = cross_val_score(rf, X, y, cv=5)
✓ 5s
print(cv_rf.mean())
... 0.7871661074908058
```

Mean CV Score = 78.72%

Cross-validation helps ensure that the model performs consistently across different subsets of the dataset.

6. Model Evaluation

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

6.1 Logistic Regression Results

Metric	Value
Accuracy	0.8162 (81.62%)
Precision	0.6804 (68.04%)

Recall	0.5764 (57.64%)
F1-Score	0.6241 (62.41%)
ROC-AUC	0.7395 (73.95%)

6.2 Random Forest Results

Metric	Value
Accuracy	0.7956 (79.56%)
Precision	0.6592 (65.92%)
Recall	0.4718 (47.18%)
F1-Score	0.5500 (55.00%)
ROC-AUC	0.6920 (69.20%)

```
[20] print("Accuracy:", accuracy_score(y_test, lr_pred))
print("Precision:", precision_score(y_test, lr_pred))
print("Recall:", recall_score(y_test, lr_pred))
print("F1:", f1_score(y_test, lr_pred))
print("ROC-AUC:", roc_auc_score(y_test, lr_pred))

Accuracy: 0.8161816891412349
Precision: 0.680379746835443
Recall: 0.5764075067024129
F1: 0.6240928882438317
ROC-AUC: 0.7394585796060328

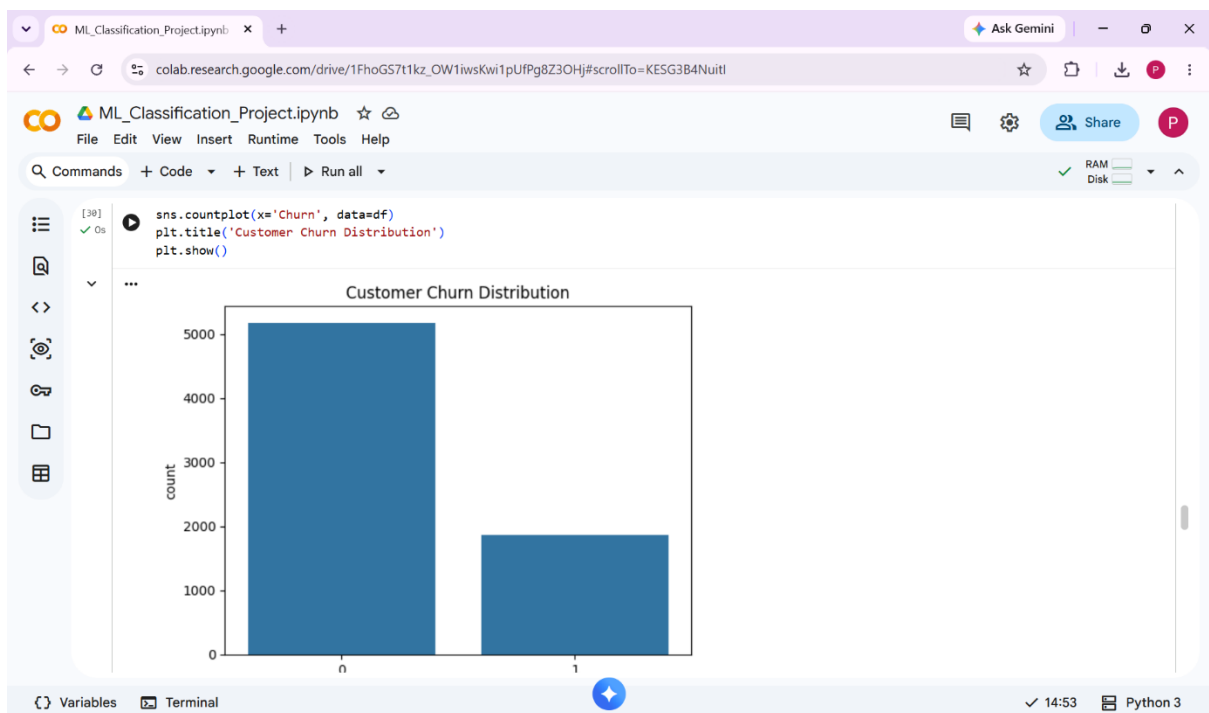
[21] print("Accuracy:", accuracy_score(y_test, rf_pred))
print("Precision:", precision_score(y_test, rf_pred))
print("Recall:", recall_score(y_test, rf_pred))
print("F1:", f1_score(y_test, rf_pred))
print("ROC-AUC:", roc_auc_score(y_test, rf_pred))

Accuracy: 0.7955997161107168
Precision: 0.6591760299625468
Recall: 0.4718498659517426
F1: 0.55
ROC-AUC: 0.6920060140569524
```

7. Model Comparison

Metric	Logistic Regression	Random Forest
Accuracy	81.62%	79.56%
Precision	68.04%	65.92%
Recall	57.64%	47.18%
F1-Score	62.41%	55.00%
ROC-AUC	73.95%	69.20%
Cross Validation Score	80.25%	78.72%

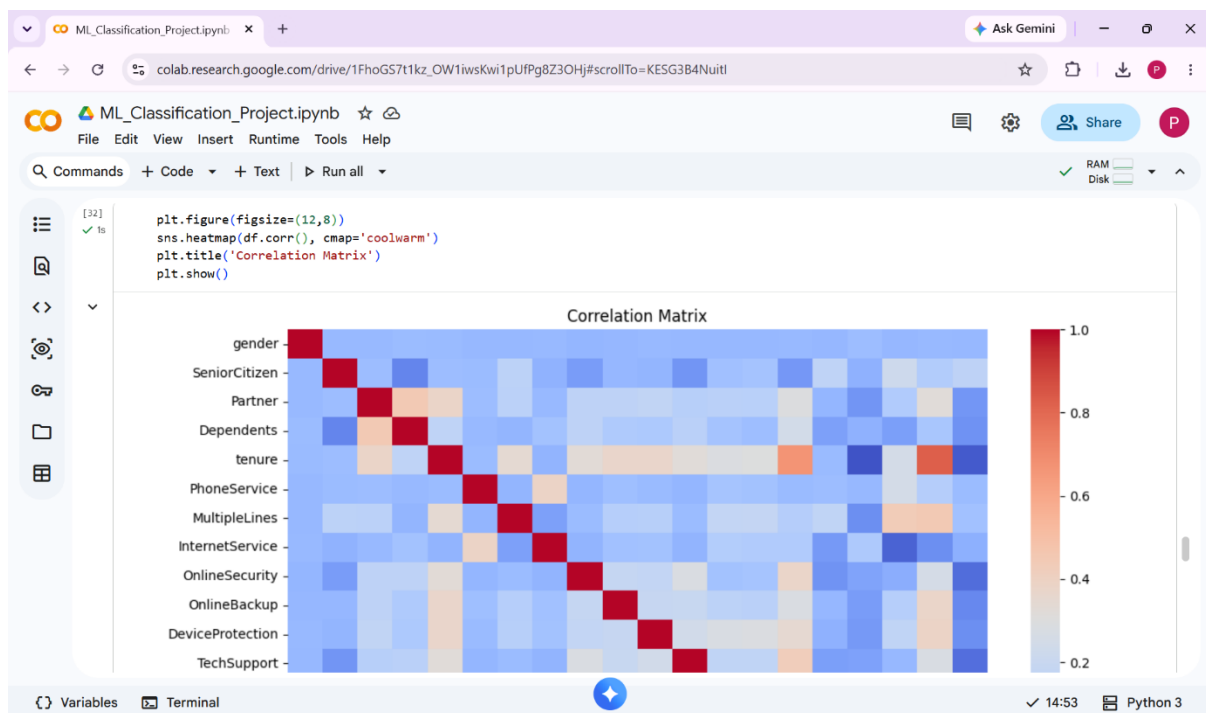
Visualizations



Analysis

The performance of both algorithms was compared using standard classification metrics. The model with higher Accuracy, F1-Score, and ROC-AUC score was considered the better-performing model.

Based on the obtained results, Logistic Regression achieved better overall performance than Random Forest Classifier because it obtained higher Accuracy, Recall, F1-Score, ROC-AUC, and Cross-Validation scores.



8. Conclusion

In this project, a machine learning classification system was developed to predict customer churn using the Telco Customer Churn dataset.

The following tasks were successfully completed:

- Data preprocessing
- Missing value handling
- Label encoding
- Train-test splitting
- Cross-validation
- Logistic Regression implementation
- Random Forest implementation
- Model evaluation and comparison

Among the evaluated models, Logistic Regression produced the best overall performance based on Accuracy, F1-Score, ROC-AUC, and Cross-Validation results.

This project demonstrates how machine learning techniques can be applied to predict customer behaviour and support business decision-making.

9. Tools and Libraries Used

Programming Language

- Python

Development Environment

- Google Colab

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

10. References

1. Kaggle – Telco Customer Churn Dataset
2. Scikit-Learn Documentation
3. Python Documentation